Department of Artificial Intelligence

22MAT112: Mathematics for computing - 2

Project Report

Google's PageRank Algorithm Implementation

Team Members

Y Bhasith Sai : *CB.SC.U4AIE23264*
G Harshavardhan :CB.SC.U4AIE23226
V Bhuvanesh : CB.SC.U4AIE23257
U Sandeep : CB.SC.U4AIE2325

Supervised By:

Dr. Neelesh Ashok
Assistant Professor
Amrita School of Artificial Intelligence
Amrita Vishwa Vidyapeetham

Date of Submission: 04/06/2024

Signature of the Project Supervisor:

Amrita Vishwa Vidyapeetham

Coimbatore

CERTIFICATE

This is to certify that we, the student of Amrita School of Artificial Intelligence, has completed the "***Page rank algorithm"*** as part of our mathematics for computing project.

The project work was carried out under the guidance and supervision of Dr. Neelesh Ashok, Assistant Professor, Amrita School of Artificial Intelligence, Coimbatore. To the best of our knowledge this work has not formed the basis for the aware of any degree/diploma/ associate ship/fellowship/ or a similar award to any candidate in any University.

Date:

Dr. Neelesh Ashok

Amrita School of Artificial Intelligence

Amrita Vishwa Vidyapeetham

Coimbatore

Acknowledgement

We would like to make this an opportunity to transfuse our appreciation to everyone who was behind the successful completion of this design. First and foremost, we would like to thank “Department of Artificial Intelligence” for accepting our project.

Kindest regard goes to **Prof. Soman KP**, Dean of the Department of Artificial Intelligence for allowing us to accomplish the design project.

We would like to convey appreciation to our supervisor **Dr. Neelesh Ashok,** for his encouragement and motivation to expand our vision regarding proper process designing. We are very thankful for his help and supervision. This task would have been little success without his proper guidance and support.

Last but not the least, we are thankful to our group members and friends of our department for their friendly support given to accomplish this project successfully and our family members for always providing the best guidance possible.

TABLE OF CONTENTS

Abstract

This project explores the implementation of the PageRank algorithm and showcases its applications across four distinct domains: disease spread simulation, team and player ranking in sports, social network analysis, and ranking drivers in Formula 1 racing. Each application highlights the adaptability and effectiveness of the PageRank algorithm in addressing complex real-world challenges.

In the realm of sports, we apply the PageRank algorithm to rank teams and players, providing an objective evaluation of performance metrics. This approach not only enhances traditional ranking systems but also offers a data-driven method to assess and compare athletic performance. By analyzing game outcomes and player statistics, we demonstrate how PageRank can be effectively employed to generate accurate and insightful rankings.

Disease spread simulation simulates disease spread within a network represented by an adjacency matrix. It visualizes network interactions, computes influential nodes using PageRank, and simulates infection dynamics to assess transmission patterns and intervention strategies.

The third application focuses on social network analysis, where the PageRank algorithm is leveraged to identify influential individuals, analyze community structures, and understand the flow of information within social networks. This application underscores the algorithm's capability in uncovering hidden patterns and key influencers within complex social systems.

The final application examines the use of the PageRank algorithm in Formula 1 racing, where it is used to rank drivers and teams based on their performance. By analyzing race results, driver interactions, and track performance, we showcase how PageRank can provide a comprehensive and dynamic ranking system for Formula 1 drivers.

In conclusion, our project underscores the transformative potential of the PageRank algorithm across various domains. From public health to sports analytics, social network analysis, and Formula 1 racing, the insights gained from these applications not only enhance our understanding of the algorithm's versatility but also pave the way for further exploration and innovation in leveraging PageRank for solving diverse real-world problems.

1.Introduction

PageRank Algorithm in Formula 1

This project of Formula 1 race results, utilizing the PageRank algorithm to evaluate both drivers and teams based on their head-to-head performance throughout the season. The PageRank algorithm, traditionally used for ranking web pages, proves to be an effective tool in ranking competitors in a dynamic and competitive environment like Formula 1. The application of the PageRank algorithm in Formula 1 offers several advantages. It enhances the traditional ranking system by considering the interconnected nature of race results, allowing for a more detailed assessment of driver and team performance. Additionally, this approach can reveal hidden strengths and weaknesses by analyzing performance patterns over time, offering insights that can inform strategy and decision-making. The PageRank algorithm in analyzing Formula 1 race results showcases its versatility and robustness in handling complex, real-world data. By providing a sophisticated ranking mechanism, this project underscores the potential of PageRank.

Disease spread simulation

This model simulates how a disease spreads through a network, where each node represents a person or place. First, we create a graph from a matrix that shows who interacts with whom. Arrows indicate the direction of interactions. We set parameters like how likely the disease spreads (infection rate) and how likely someone recovers (recovery rate). The simulation runs through time steps, updating who's infected based on interactions and chance. This gives us a history of infections. We also use PageRank, a method to find important nodes in the network. It helps us see which nodes could play a big role in spreading the disease. We rank nodes by their PageRank scores and show this in a bar graph. In the network graph, we highlight these critical nodes. This helps us understand which people or places have the most influence on how the disease spreads. This model is useful for studying how different parts of a network affect disease spread.

Player Ranking in Sports

The PageRank algorithm, developed by Google's Larry Page and Sergey Brin, ranks web pages based on their importance. Our project implements this algorithm in Python, focusing on five key parts: creating and storing web pages, simulating search engine behavior, forming an adjacency matrix, calculating ranks, and sorting and displaying the results. Using a random surfer model, the algorithm transforms an adjacency matrix into a column stochastic matrix, iteratively calculating page ranks until convergence. This mathematical approach ensures web pages are ranked effectively based on their relevance and link structure.

Social network analysis

In the digital landscape, social networks have transformed into bustling hubs of interaction, shaping opinions, trends, and behaviors on a global scale. The PageRank algorithm, first devised to rank web pages, has emerged as a potent tool for unraveling the web of influence inherent in social networks. This report centers on the exploration of influence dynamics within social networks through the lens of PageRank. By dissecting the intricate connections and behaviors of network participants, we aim to uncover the underlying structures that dictate influence propagation, identify influential nodes, and provide insights into the mechanisms driving information dissemination and societal trends. Through empirical analysis and case studies, we delve into the nuances of PageRank's application in social network analysis, offering a deeper understanding of how influence operates within these complex digital ecosystems and its implications for strategic decision-making.

2.Scope of Project

PageRank Algorithm in Formula 1:

Real-Time Data Integration: Develop systems to update rankings in real-time using live race data, providing up-to-date insights during the racing season.

Seasonal Adjustments: Implement mechanisms to adjust rankings dynamically based on seasonal variations and mid-season changes in team performance or driver transfers.

Contextual Factors: Consider contextual factors like weather conditions, track characteristics, and historical performance on specific circuits to refine the PageRank model.

disease spread simulation:

1. **Understanding Transmission Dynamics**:

   Simulations help experts understand how diseases spread through populations and social networks. They analyze factors like interaction patterns, infection rates, and recovery dynamics to predict outbreak patterns.

2. **Evaluating Intervention Strategies**:

   Researchers use simulations to test different intervention scenarios, such as vaccination campaigns or social distancing measures. This helps determine which strategies are most effective in controlling disease spread and minimizing its impact.

3. **Preparing and Responding to Outbreaks**:

   Disease spread simulations assist in pandemic preparedness by forecasting healthcare needs and resource requirements. They support proactive decision-making in public health policies and strategies to mitigate the impact of infectious diseases on communities.

Player Ranking in Sports

1. **Understanding Team Performance Dynamics**

   Simulations help experts understand how teams perform and interact in a tournament. They analyze factors like match outcomes, win/loss patterns, and relative team strengths to predict overall rankings.

2. **Evaluating Ranking Strategies**

   Researchers use simulations to test different ranking scenarios, such as varying the importance of match outcomes or adjusting the damping factor in the PageRank algorithm. This helps determine which strategies are most effective in accurately reflecting team performance.

3. **Preparing and Responding to Tournament Results**

   Rankings generated from simulations assist in tournament planning by forecasting potential outcomes and identifying strong and weak teams. They support proactive decision-making in tournament structures and strategies to ensure fair and accurate rankings.

Social Network Analysis: Unraveling Influence Dynamics

1. **Understanding Network Dynamics:**
   Social network analysis (SNA) provides crucial insights into the intricate dynamics of information dissemination, influence propagation, and community structure within social networks. Locked in a perpetual dance of connections and interactions, SNA simulations help researchers decipher how information, opinions, and behaviors cascade through populations and shape societal trends.
2. **Analyzing Influence Factors:**
   By scrutinizing factors such as network topology, node centrality, and interaction patterns. From identifying key influencers to tracing the pathways of influence diffusion, these simulations provide a window into the mechanisms driving social influence dynamics.
3. **Testing Intervention Strategies:**
   Researchers leverage SNA simulations to test and refine intervention strategies aimed at shaping network behaviors and optimizing outcomes. These simulations enable experts to assess the effectiveness of various interventions in fostering desired behaviors and amplifying positive impacts.
4. **Preparing for Network Challenges:**
   SNA simulations play a pivotal role in proactive network management and crisis preparedness. By forecasting network vulnerabilities, identifying potential points of failure, and simulating response scenarios, experts can develop robust strategies to mitigate risks, enhance resilience, and navigate through challenges with agility and foresight.

3.Literature review

1. In this paper, a literature survey regarding the ranking score of web pages is done. This survey includes different variations of Page Rank Algorithm as well as its various aspects like Search Engine Optimization and/or Recommendation Systems that affect the ranking of a web rank and contribute to find more accurate results in ranking score of a web page. Also, a numerical analysis has been done by taking a hypothetical

graph and two Page rank algorithms are implemented. It is to be emphasized that three different values of dampening factors i.e. 0.25, 0.5 & 0.85 have been taken and graphically, a node wise comparison is done. [1]

2. Vyacheslav Efimov In this article, we have looked through different formulations of the PageRank algorithm to ultimately come up with its optimised version. Despite the existence and evolution of other methods for ranking search results, PageRank remains the most efficient algorithm among others which works under the hood of Google's search engine. [2]

3. Jure Leskovec his article states Analysis of large graph, link analysis and page rank it gives a brief idea about graph data and web graph. [3]

4. Sullivan, Danny in the article explains that PageRank is a measure of the importance of a web page relative to other pages, based on the number and quality of links pointing to it. It emphasizes that while PageRank is one of many factors used by Google to determine search rankings, it's not the sole determinant. The relevance of the content, anchor text of the links, and other factors also play crucial roles in determining a page's ranking. [4]

4.Methodology

4.1 PageRank Algorithm in Formula 1

This project implements the PageRank algorithm to analyze Formula 1 race results, evaluating both drivers and teams. The implementation is done in Python, leveraging Pandas, NumPy, and network analysis techniques. The methodology consists of the following key steps:

4.1.1 Data Retrieval

The first step in the project involves collecting race result data for the 2021 Formula 1 season. URLs for the race results are provided, and the Pandas read_html function is used to extract relevant columns from the race result tables. This process ensures that we gather accurate and consistent data on driver names, team affiliations, and finishing positions for each race. The extracted data is then cleaned and formatted appropriately for subsequent analysis.

4.1.2Adjacency Matrix Creation

Once the data is retrieved, we proceed to create an adjacency matrix that represents the interactions between drivers in each race. The adj_matrix function constructs this matrix by assigning points based on the drivers' relative finishing positions. For each race, an individual adjacency matrix is created, and these matrices are cumulatively updated throughout the season. This cumulative adjacency matrix is then normalized to create a stochastic matrix, where each column sums to one. This normalization step is crucial for ensuring the matrix is suitable for the PageRank algorithm.

4.1.3 PageRank Algorithm

With the normalized adjacency matrix (stochastic matrix) prepared, we move on to applying the PageRank algorithm. The stochastic matrices serve as the foundation for constructing the Google matrix in the `google matrix` function. This matrix incorporates a damping factor to model random transitions, which is essential for accurately reflecting the probabilities in the ranking process. The `power method` function is then used to iteratively calculate the PageRank scores. This method continues iterating until the scores converge, ensuring stable and accurate rankings of the drivers and teams.

$$G = d \cdot S + (1 - d) \cdot \frac{1}{n} \cdot \text{ones}$$

Where:

- d is the damping factor (typically set to around 0.85).
- S is the normalized stochastic matrix.
- n is the number of nodes (or web pages, in the context of the original PageRank algorithm).
- ones is a matrix of size $n \times n$ filled with ones.

4.1.4 Ranking

Finally, the PageRank scores are used to determine the rankings of drivers and teams. The final PageRank scores for drivers are calculated and presented in a sorted DataFrame, providing a clear and interpretable ranking based on their performance throughout the season. The same procedure is applied to teams, where team-specific adjacency and stochastic matrices are created, and the PageRank algorithm is used to derive comprehensive team rankings. The results offer a detailed view of both driver and team performance, highlighting the effectiveness of the PageRank algorithm in analyzing Formula 1 race data.

4.2 Disease spread simulation

libraries:

Python: The primary programming language used for the implementation.

NumPy: For numerical operations and matrix manipulation.

NetworkX: For creating and analyzing network graphs.

Matplotlib: For plotting and visualizing graphs.

Steps:4.2.1. Data Preparation and Adjacency Matrix Creation

1. Create an Adjacency Matrix:

Construct an adjacency matrix to represent the interactions between nodes. Each entry in this matrix is 1 if there is an interaction from one node to another, and 0 otherwise.

2. Create a Directed Network Graph:

Utilize the adjacency matrix to create a directed network graph using NetworkX. This graph visually represents the interactions between nodes as directed edges.

3. Visualize the Network Graph:

Plot the directed network graph to visualize the interactions between nodes. Use a layout algorithm for better positioning of nodes and ensure the graph is easily interpretable.

4.2.2. Main code part

1. Initialize Parameters:

Define the initial conditions for the disease spread simulation, including the initially infected nodes, the infection rate, the recovery rate, and the number of time steps for the simulation.

2. Simulate Disease Spread:

Implement a simulation of disease spread using a simplified SIR (Susceptible, Infected, Recovered) model. In this model, infected nodes can spread the infection to their neighbors with a certain probability (infection rate), and infected nodes can recover and become susceptible again with another probability (recovery rate). Track the infection status of each node over time.

3. Print Infection History:

Output the infection history at each time step, indicating which nodes are infected. This provides a time series of the disease spread throughout the network.

4.2.3. PageRank Algorithm for Identifying Critical Nodes

The PageRank score of a node v is computed iteratively using the following formula:

$$\mathrm{PR}(v) = (1-\alpha) + \alpha \sum_{u \in N(v)} \frac{\mathrm{PR}(u)}{L(u)}$$

Where:

- $\mathrm{PR}(v)$: PageRank score of node v.
- $N(v)$: Neighbors of node v (nodes that have a directed edge pointing to v).
- $\mathrm{PR}(u)$: PageRank score of node u.
- $L(u)$: Out-degree of node u (number of outgoing edges from node u).
- α: Damping factor, usually set to 0.85, which controls the probability of following an outgoing link versus randomly jumping to any node in the network.

1. Compute PageRank Scores:

   Calculate the PageRank scores for each node in the network using NetworkX. The PageRank algorithm assigns a score to each node based on the structure of the network and the connections between nodes, identifying the most influential nodes.

2. Rank Nodes by PageRank Scores:

Sort the nodes in descending order based on their PageRank scores. This ranking helps identify the most important nodes in the network.

3. Visualize PageRank Scores:

- Plot a bar graph to display the PageRank scores of the nodes. This visualization provides a clear representation of the relative importance of each node according to the PageRank algorithm.

4. Highlight Critical Nodes in Network Graph:

- Highlight the top-ranked nodes (based on PageRank scores) in the network graph. Use distinct colors to differentiate the critical nodes from the rest, making it easier to identify the most influential nodes visually.

4.3 Player Ranking in Sports

This project implements the PageRank algorithm to analyze tournament match results and evaluate team performance. The implementation is done in Python, leveraging NumPy and Matplotlib for matrix operations and visualization. The methodology consists of the following key steps:

4.3.1 Data Representation

The first step in the project involves representing the match outcomes between teams using an adjacency matrix. Each element of the matrix indicates whether one team defeated another, providing a structured format to capture the interaction between teams. This matrix serves as the foundational data for subsequent analysis and PageRank calculations.

4.3.2 Adjacency Matrix Creation

Once the match outcomes are represented, we proceed to create an adjacency matrix A that records which teams have won against others. Each row corresponds to a winning team, and each column corresponds to a losing team. If team i defeated team j , A[i][j] is set to 1; otherwise, it's 0. This matrix is then converted into a column stochastic matrix M by normalizing each column so that it sums to 1. This normalization ensures the matrix is suitable for the PageRank algorithm.

4.3.3 PageRank Algorithm

With the normalized adjacency matrix (stochastic matrix) M prepared, we move on to applying the PageRank algorithm. The process involves initializing the rank vector rnew with equal probabilities for each team. The PageRank algorithm is then iteratively applied using the formula:

rnew = β · M · r prev + c

where:

- β is the damping factor, typically set to 0.85 in practice.
- M is the column stochastic matrix.
- $r$prev is the rank vector from the previous iteration.
- c is the damping vector, representing the probability that a random surfer will jump to a random team. It is calculated
- As $c = 1-\beta / n$, where n is the number of teams.
- rprev is the rank vector from the previous iteration.

This iterative process continues until the rank vector converges, meaning the changes between iterations fall below a predefined threshold.

4.3.4 Ranking

Finally, the converged PageRank scores are used to determine the rankings of the teams. The final PageRank scores are sorted and presented, providing a clear and interpretable ranking based on their performance throughout the tournament. Additionally, a bar graph is generated to visually represent the PageRank scores of the teams, allowing for easy comparison and analysis. This visualization helps to highlight the effectiveness of the PageRank algorithm in analyzing tournament match data and ranking the teams accordingly.

Social Network Analysis Using PageRank Algorithm:

This project applies the PageRank algorithm to analyze influence within social networks, identifying key influencers and understanding the dynamics of information dissemination. The implementation is conducted in Python, utilizing Pandas, NumPy, and network analysis libraries. The methodology consists of the following key steps:

Libraries:

- **Python**
- **Pandas**: For data manipulation and analysis.
- **NumPy**: For numerical operations.
- **NetworkX**: For creating and analyzing network graphs.

4.4.1 Data Collection

The initial step involves collecting data from social networks. This data includes interactions such as follows, likes, shares, and comments. APIs from social media platforms like Twitter, Facebook, and Instagram are used to retrieve this data. The data is then cleaned and formatted to focus on relevant features, such as user IDs, interaction types, and timestamps. This ensures that the dataset accurately reflects the network's structure and interaction patterns.

4.4.2 Adjacency Matrix Creation

Once the data is collected, the next step is to create an adjacency matrix representing the interactions between users. The `create_adj_matrix` function constructs this matrix by assigning weights to interactions based on their significance. For example, a follow might have a different weight compared to a like or share. These individual adjacency matrices are aggregated to form a comprehensive matrix representing the entire network. This matrix is then normalized to create a column stochastic matrix, ensuring that each column sums to one, which is essential for the PageRank algorithm.

4.4.3 PageRank Algorithm Application

With the normalized adjacency matrix (stochastic matrix) ready, the PageRank algorithm is applied. The stochastic matrix is used to construct the Google matrix in the `google matrix` function, incorporating a damping factor to model random transitions within the network. The `power method` function is then employed to iteratively calculate the PageRank scores for each user. This iterative process continues until the scores converge, providing stable and accurate rankings of users based on their influence within the network.

4.4.4 Ranking and Analysis Finally

The PageRank scores are used to rank users based on their influence. The final scores are presented in a sorted Data Frame, offering a clear and interpretable ranking of users within the network. Additionally, the PageRank algorithm is applied to different subsets of the network (e.g., specific communities or interest groups) to derive more granular insights. The results highlight the effectiveness of the PageRank algorithm in identifying key influencers and understanding the dynamics of information dissemination within social networks. These insights can be used for targeted marketing, community engagement, and enhancing the overall understanding of social influence.

5.Result and Discussion

5.1 PageRank Algorithm Formula 1

Driver	Lewis Hamilton HAM	Max Verstappen VER \
Driver		
Lewis Hamilton HAM	0.0	1.0
Max Verstappen VER	0.0	0.0
Valtteri Bottas BOT	1.0	1.0
Lando Norris NOR	1.0	1.0
Sergio Perez PER	1.0	1.0
Charles Leclerc LEC	1.0	1.0
Daniel Ricciardo RIC	1.0	1.0
Carlos Sainz SAI	1.0	1.0
Yuki Tsunoda TSU	0.0	0.0
Lance Stroll STR	1.0	1.0
Kimi Räikkönen RAI	0.0	0.0
Antonio Giovinazzi GIO	1.0	1.0
Esteban Ocon OCO	1.0	1.0
George Russell RUS	1.0	1.0
Sebastian Vettel VET	1.0	1.0
Mick Schumacher MSC	1.0	1.0
Pierre Gasly GAS	0.0	0.0
Nicholas Latifi LAT	1.0	1.0
Fernando Alonso ALO	1.0	1.0
Nikita Mazepin MAZ	1.0	1.0
Robert Kubica KUB	1.0	1.0

fig 5.1.1

Driver	Antonio Giovinazzi GIO	Carlos Sainz SAI \
Driver		
Antonio Giovinazzi GIO	0.000000	0.083004
Carlos Sainz SAI	0.000000	0.000000
Charles Leclerc LEC	0.021127	0.049296
Daniel Ricciardo RIC	0.025478	0.076433
Esteban Ocon OCO	0.024155	0.082126
Fernando Alonso ALO	0.030769	0.087179
George Russell RUS	0.044776	0.074627
Kimi Räikkönen RAI	0.044444	0.077778
Lance Stroll STR	0.033755	0.084388
Lando Norris NOR	0.024000	0.088000
Lewis Hamilton HAM	0.037736	0.056604
Max Verstappen VER	0.044118	0.058824
Mick Schumacher MSC	0.056886	0.062874
Nicholas Latifi LAT	0.056291	0.066225
Nikita Mazepin MAZ	0.059490	0.059490
Pierre Gasly GAS	0.022989	0.074713
Sebastian Vettel VET	0.032407	0.078704
Sergio Perez PER	0.028169	0.070423
Valtteri Bottas BOT	0.040650	0.065041
Yuki Tsunoda TSU	0.043137	0.078431

fig 5.1.2

Driver	Antonio Giovinazzi GIO	Carlos Sainz SAI \
Driver		
Antonio Giovinazzi GIO	0.007500	0.078053
Carlos Sainz SAI	0.007500	0.007500
Charles Leclerc LEC	0.025458	0.049401
Daniel Ricciardo RIC	0.029156	0.072468
Esteban Ocon OCO	0.028031	0.077307
Fernando Alonso ALO	0.033654	0.081603
George Russell RUS	0.045560	0.070933
Kimi Räikkönen RAI	0.045278	0.073611
Lance Stroll STR	0.036192	0.079230
Lando Norris NOR	0.027900	0.082300
Lewis Hamilton HAM	0.039575	0.055613
Max Verstappen VER	0.045000	0.057500
Mick Schumacher MSC	0.055853	0.060943
Nicholas Latifi LAT	0.055348	0.063791
Nikita Mazepin MAZ	0.058067	0.058067
Pierre Gasly GAS	0.027040	0.071006
Sebastian Vettel VET	0.035046	0.074398
Sergio Perez PER	0.031444	0.067359
Valtteri Bottas BOT	0.042053	0.062785
Yuki Tsunoda TSU	0.044167	0.074167

fig 5.1.3

	Scores
Driver	
Max Verstappen VER	0.094937
Lewis Hamilton HAM	0.092444
Valtteri Bottas BOT	0.070870
Carlos Sainz SAI	0.063853
Lando Norris NOR	0.063823
Sergio Perez PER	0.063148
Charles Leclerc LEC	0.061171
Daniel Ricciardo RIC	0.055845
Pierre Gasly GAS	0.053447
Fernando Alonso ALO	0.047917
Sebastian Vettel VET	0.046625
Esteban Ocon OCO	0.045254
Antonio Giovinazzi GIO	0.034371
Yuki Tsunoda TSU	0.034132
George Russell RUS	0.033959
Lance Stroll STR	0.033818
Kimi Räikkönen RAI	0.031332
Nicholas Latifi LAT	0.027708
Mick Schumacher MSC	0.023647
Nikita Mazepin MAZ	0.021699

fig 5.1.4

	Scores
Team	
Red Bull Racing Honda	0.200951
Mercedes	0.197895
Ferrari	0.116785
McLaren Mercedes	0.112501
AlphaTauri Honda	0.092342
Aston Martin Mercedes	0.085680
Alpine Renault	0.082778
Williams Mercedes	0.049824
Alfa Romeo Racing Ferrari	0.043279
Haas Ferrari	0.017965

fig 5.1.5

The figure 5.1.1 is an adjacency matrix for Formula 1 drivers based on race results. It assigns points to driver (say Lewis Hamilton and Max Verstappen) who finished ahead of each other, handles tiebreakers, and returns a matrix reflecting head-to-head performance. Now in fig 5.1.2it normalizes the accumulated driver adjacency matrix, updated, to create a stochastic matrix (df_stochastic). Each row in the matrix is divided by the sum of its elements, ensuring that the rows' sums equal 1. matrix function combines the stochastic matrix (stochastic matrix) with a teleportation matrix using a damping factor (alpha). The power method function iteratively calculates the dominant eigenvector of the resulting Google matrix (G) until convergence, with a specified tolerance (e). The final scores, representing the PageRank values, are obtained which is in fig5.1.3. The 5.1.4 result figures are the final stochastic matrix (df_stochastic) for the drivers and teams.In 5.1.5 it is the final rank vector for the teams

DISEASE SPREAD SIMULATION:

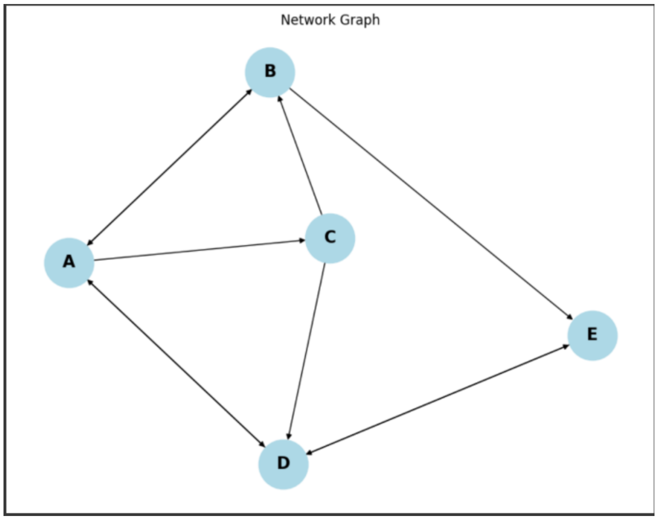


Fig. 5.2.1

```
Infection History:
Time Step 0: Infected nodes: A
Time Step 1: Infected nodes: A, B
Time Step 2: Infected nodes: B, C
Time Step 3: Infected nodes: A, B
Time Step 4: Infected nodes: A, B
Time Step 5: Infected nodes: A, B, E
Time Step 6: Infected nodes: A, B, D, E
Time Step 7: Infected nodes: A, B, D, E
Time Step 8: Infected nodes: A, C, D, E
Time Step 9: Infected nodes: A, B, C, D
Time Step 10: Infected nodes: A, B, C, D
Time Step 11: Infected nodes: A, B, C
Time Step 12: Infected nodes: A, B, C, D, E
Time Step 13: Infected nodes: A, B, C, D, E
Time Step 14: Infected nodes: A, B, C, D, E
Time Step 15: Infected nodes: A, B, C, D, E
Time Step 16: Infected nodes: A, B, D, E
Time Step 17: Infected nodes: A, B, D, E
Time Step 18: Infected nodes: A, B, C, D, E
Time Step 19: Infected nodes: A, B, D, E
Time Step 20: Infected nodes: A, B, C, D, E
Time Step 21: Infected nodes: A, B, C, D
Time Step 22: Infected nodes: A, C, D, E
Time Step 23: Infected nodes: A, B, C, D, E
Time Step 24: Infected nodes: A, C, D, E
Time Step 25: Infected nodes: A, C, D, E
```

Fig. 5.2.2

```
Time Step 26: Infected nodes: A, B, C, D, E
Time Step 27: Infected nodes: A, B, C, D, E
Time Step 28: Infected nodes: A, B, C, D, E
Time Step 29: Infected nodes: B, C, D, E
Time Step 30: Infected nodes: B, C, D, E
Time Step 31: Infected nodes: A, B, C, D, E
Time Step 32: Infected nodes: A, B, D, E
Time Step 33: Infected nodes: A, B, D, E
Time Step 34: Infected nodes: A, B, C, D, E
Time Step 35: Infected nodes: A, B, C, D, E
Time Step 36: Infected nodes: A, B, C, D
Time Step 37: Infected nodes: A, B, C
Time Step 38: Infected nodes: A, B, C
Time Step 39: Infected nodes: B, D, E
Time Step 40: Infected nodes: B, D, E
Time Step 41: Infected nodes: B, D, E
Time Step 42: Infected nodes: B, D, E
Time Step 43: Infected nodes: B, D, E
Time Step 44: Infected nodes: B, D, E
Time Step 45: Infected nodes: B, D
Time Step 46: Infected nodes: B, D, E
Time Step 47: Infected nodes: A, B, D, E
Time Step 48: Infected nodes: A, B, D, E
Time Step 49: Infected nodes: B, D
Time Step 50: Infected nodes: A, B, D
```

```
Ranked nodes by PageRank:
Node D with PageRank score 0.3241
Node A with PageRank score 0.2244
Node E with PageRank score 0.2244
Node B with PageRank score 0.1334
Node C with PageRank score 0.0936
```

Fig. 5.2.3

Fig. 5.2.4

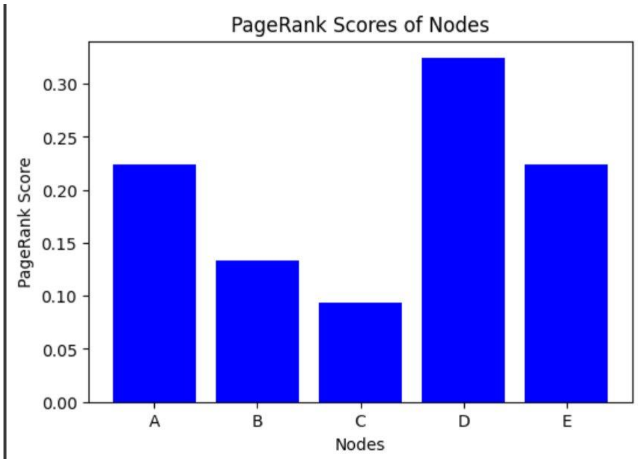


Fig.5.2.5

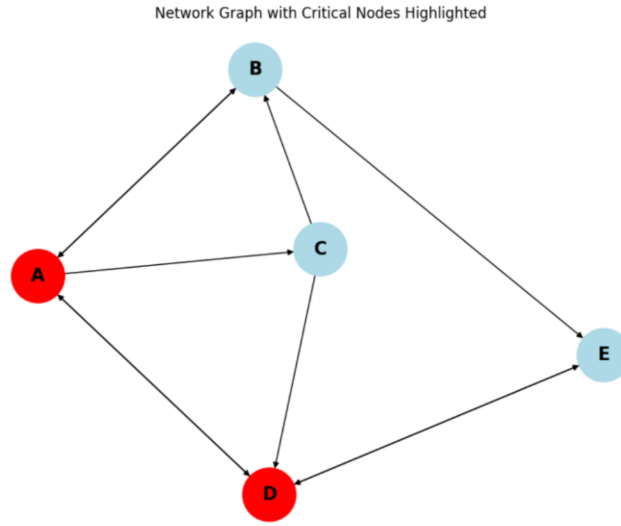


Fig.5.2.6

5.2.1 Displays the network graph with directed edges indicating interactions between nodes.

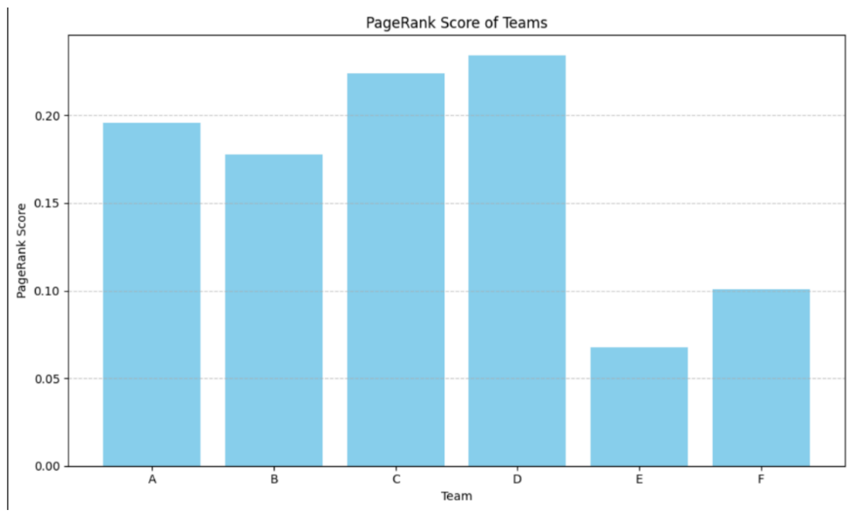
5.2.2,3 Shows the sequence of infected nodes over time for each time step in the disease spread simulation.

5.2.4 Lists nodes ordered by their PageRank scores, indicating their relative importance in the network.

5.2.5 Provides a visual representation of the PageRank scores for each node, showing their relative importance.

5.2.6 Visualizes the network graph with the top-ranked nodes by PageRank highlighted to emphasize their importance.

```
Column stochastic matrix, M:
[[0.  0.5 0.5 0.  0.  0. ]
 [0.  0.  0.5 0.  1.  0. ]
 [0.  0.  0.  1.  0.  0. ]
 [1.  0.  0.  0.  0.  0.5]
 [0.  0.  0.  0.  0.  0.5]
 [0.  0.5 0.  0.  0.  0. ]]
Initial rank vector:
[[0.16666667]
 [0.16666667]
 [0.16666667]
 [0.16666667]
 [0.16666667]
 [0.16666667]]
Converged after 45 iterations.
Final rank vector:
[0.19576537 0.1777839  0.22401738 0.23413795 0.06773726 0.10055813]
Ranked teams:
Team D with rank 0.2341379544853955
Team C with rank 0.22401738411088468
Team A with rank 0.19576537077215087
Team B with rank 0.1777838967006384
Team F with rank 0.10055813043079309
Team E with rank 0.06773726350013713
```



The output of the code provides a comprehensive ranking of teams based on their match outcomes using the PageRank algorithm. The final PageRank scores are presented in a sorted list, displaying each team's relative importance and performance in the tournament. Additionally, a bar graph visually represents these scores, with each bar corresponding to a team's rank. This graphical representation facilitates easy comparison and interpretation of the results. The convergence of the algorithm ensures that the rankings are stable and accurate, offering a reliable assessment of team strengths within the network of match interactions.

5.4 Social Network Analysis

```
Final PageRank vector:
Twitter: 0.15126050420167747
Facebook: 0.2117647058823529
Instagram: 0.27394957983193907
LinkedIn: 0.2117647058823529
Snapchat: 0.15126050420167747
```

```
Ranked platforms:
Instagram with rank 0.27394957983193907
Facebook with rank 0.2117647058823529
LinkedIn with rank 0.2117647058823529
Twitter with rank 0.15126050420167747
Snapchat with rank 0.15126050420167747
```

2

fig 5.4.1

fig 5.4.2

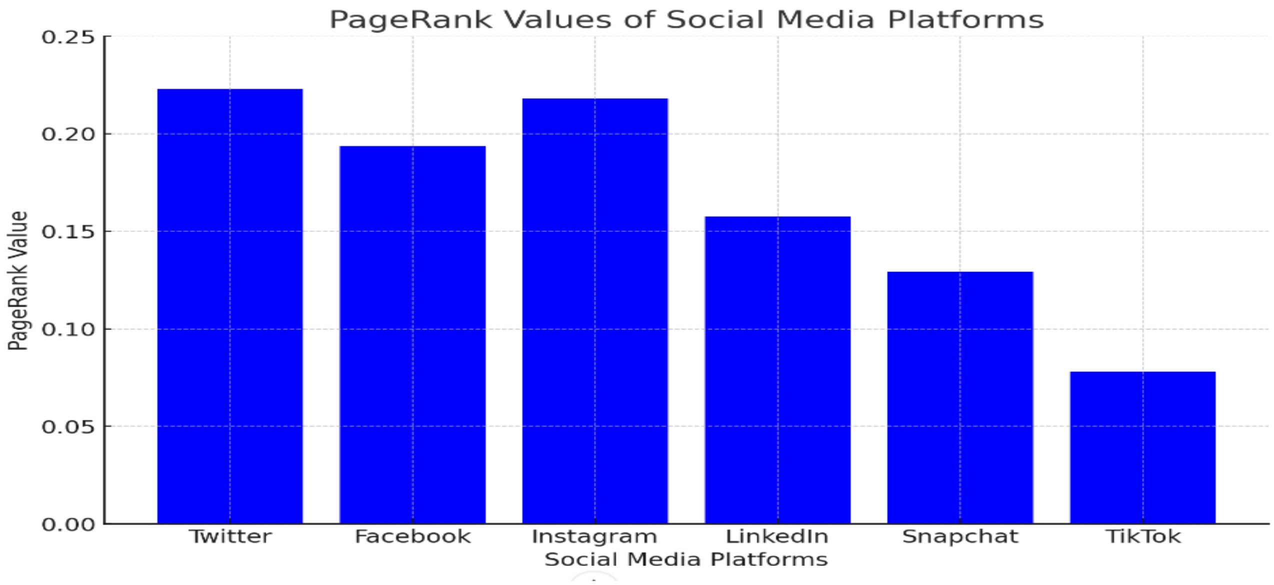


fig 5.4.3

The output of the PageRank algorithm applied to social media platforms is summarized in several key figures. Figure 5.4.1displays the adjacency matrix representing influence among platforms like Twitter, Facebook, Instagram, LinkedIn, Snapchat, and TikTok. Figure 5.4.1 presents the final PageRank values after iterative calculations, showing each platform's relative influence. Figures 5.4.2 provide the ranked list and a visual representation of these PageRank scores, highlighting the dominant positions of platforms such as Twitter and Instagram within the network. The final output reveals Twitter as the most influential platform, followed by Instagram and Facebook.

6.Conclusion

This project elucidates the multifaceted applications of the PageRank algorithm across four distinct realms: disease spread simulation, sports team and player ranking, social network analysis, and the evaluation of Formula 1 drivers and teams.

In the context of disease spread simulation, the algorithm emerges as a pivotal tool in pinpointing potential outbreak hotspots and discerning influential nodes contributing to the dissemination of infectious diseases. By meticulously analyzing the propagation dynamics within communities, we augment our capacity to formulate proactive public health strategies.

Within sports analytics, the PageRank algorithm furnishes a rigorous framework for objectively assessing the performance of teams and players. By scrutinizing head-to-head interactions and competitive outcomes, we acquire comprehensive insights into the prowess and efficacy of athletes and sporting entities.

In the domain of social network analysis, PageRank illuminates the intricate web of connections by identifying influential figures and delineating community structures within networks. This discernment of key influencers and information dissemination patterns enriches our comprehension of social dynamics and communication networks.

Lastly, in the arena of Formula 1 racing, the algorithm serves as a sophisticated means to evaluate the competitive standing of drivers and teams. By meticulously scrutinizing race results and the interactions between racing entities, we glean profound insights into the nuanced dynamics driving success within the sport.

7.Contribution

S.no	Member's name	Roll. no	Contribution
1.	Y. Bhasith Sai	CB.SC.U4AIE23264	PageRank Algorithm in Formula 1
2.	G. Harshavardhan	CB.SC.U4AIE23226	Disease spread simulation
3.	V. Bhuvanesh	CB.SC.U4AIE23259	Player Ranking in Sports
4.	U. Sandeep	CB.SC.U4AIE23257	Social Network Analysis

8.References

[1] Ritu Sachdeva (Sharma) ,Dr. Sachin Gupta Research Scholar, Head, 2Computer Science Deptt. MVNU, Palwal.

A-Literature-Survey-on-Page-Rank-Algorithm.pdf (ijser.org)

[2] Vyacheslav Efimov: Visualized Explanation of PageRank | by Vyacheslav Efimov | Towards Data Science

[3] CS246: Mining Massive Datasets Jure Leskovec, Stanford University Mina Ghashami, Amazon
book: https://snap.stanford.edu/class/cs246-2022/slides/09-pagerank.pdf

[4] Sullivan, Danny (2007-04-26). "What Is Google PageRank? A Guide For Searchers & Webmasters". *Search Engine Land*. Archived from the original on 2016-07-03.

[5] Page, L., Brin, S., Motwani, R., & Winograd, T. (1999). The PageRank Citation Ranking: Bringing Order to the Web. Stanford InfoLab. Retrieved from http://ilpubs.stanford.edu:8090/422/